

# Md Ashraful Hossen Akash

Rajshahi, Bangladesh | ✉ [ashraful.hossen.akash@gmail.com](mailto:ashraful.hossen.akash@gmail.com)

🌐 [LinkedIn](#) | 🌐 [Website](#) | ☎ (+880)1764503270

## RESEARCH INTERESTS

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Computer Vision, Large Language Model, Natural Language Processing.

## EDUCATION

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**B.Sc. in Computer Science & Engineering**

Mar 2022 - Jun 2026

Rajshahi University of Engineering & Technology

**Result (CGPA): 3.17/4.00 (3.61 is last 3 semesters)**

## PUBLICATIONS

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- Md. Asif Sajeed, Shyla Afroge, **Md. Ashraful Hossen Akash**, Abul Khayer, Abdullah Al Mamun. “Memory-Efficient Deep Learning Architectures for Water Body Segmentation in Satellite Imagery: A Comparative Analysis and Ensemble.” **(Accepted)**  
*Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT 2025).* [\[ResearchGate\]](#)
- Abul Khayer, Shyla Afroge, Abdullah Al Mamun, Md. Asif Sajeed **Md. Ashraful Hossen Akash**, et al. “Transformer-Based Multi-Label Disease Prediction from Radiology Reports.” **(Accepted)**  
*Proceedings of the 28th International Conference on Computer and Information Technology (ICCIT 2025).* [\[ResearchGate\]](#)
- **Md. Ashraful Hossen Akash**. “An Explainable Deep Learning Framework for Multi-Class Alzheimer’s Diagnosis with Multi-Scale CNN.” **(Submitted)**  
*Proceedings of 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026).*

## EXPERIENCE

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**Research Assistant (Under Shyla Afroge)**

Dec 2024 – Present

- Assisted in deep learning and NLP research, including data preprocessing, model implementation, and experimental evaluation.
- Contributed to conference paper preparation, result analysis, and revisions.

**Software Engineer Intern**

Feb 2025 – May 2025

Code Studio Software Company

*Rajshahi, Bangladesh*

- Gained practical exposure to software company structure, recruitment processes, and the Software Development Life Cycle (SDLC).
- Designed and implemented real-world database schemas; practiced SQL queries including CRUD operations, JOINS, and aggregate functions.
- Learned backend development concepts such as routing, middleware, authentication tokens, REST API development, and rate limiting.
- Developed a full-stack web application (CRUD / Blog system) using modern frameworks such as Node.js, Django, React, and Next.js with MySQL / MongoDB / PostgreSQL.
- Enhanced professional skills through job preparation tasks, CV writing, and higher study presentations; improved GitHub and LinkedIn profiles.

## EDGE Training Program (PHP – Laravel)

Bangladesh Computer Council (ICT Division)

May 2024 – Feb 2025

CSE, RUET

- Completed an 8-month professional training under the “Enhancing Digital Government and Economy (EDGE)” project.
- Gained hands-on experience in PHP and Laravel for backend web application development.
- Developed CRUD-based web applications following MVC architecture and best practices.
- Acquired practical knowledge of database integration, authentication, and RESTful APIs.

## Private Tutor (Mathematics, Physics, Programming)

Mar 2022 – Present

- Tutored secondary and higher-secondary students in Mathematics, Physics, ICT, C programming, HTML, and database fundamentals.
- Explained core concepts using problem-solving techniques and practical examples.
- Assisted students in exam preparation, assignments, and foundational programming skills.

## COMPETITIONS

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### Kaggle Competitions

- **Santa 2025 - Christmas Tree Packing** (Ongoing) — Current Rank: **276 / 2653 (Top 10.4%)**.
- **Predicting Student Test Scores** (Ongoing) — Current Rank: **83 / 1304 (Top 6.4%)**.
- Applied feature engineering, model training, and ensemble techniques to improve leaderboard performance on real-world tabular datasets.
- Kaggle profile: [Ashraful](#)

### Codeforces Competitions

- Active participant in Codeforces rated contests (Div. 2, Div. 3, Div. 4).
- Current rating: **995**, demonstrating consistent improvement through regular contest participation.
- Solved algorithmic problems involving arrays, greedy techniques, data structures, basic dynamic programming, and graph traversal.
- Codeforces profile: [ash\\_ra\\_ful](#)

## PROJECTS

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### AlzDetect-Ai: Explainable Alzheimer’s Disease Classification System

Developed an AI-powered system for multi-class Alzheimer’s disease diagnosis from brain MRI scans. Implemented a custom multi-scale CNN and an ensemble of ResNet50 and Xception models for robust classification across four disease stages.

- Achieved high multi-class performance on 33,000+ MRI images with strong accuracy, precision, recall, and AUC scores.
- Integrated explainable AI techniques including Grad-CAM, Integrated Gradients, and LIME to visualize discriminative neuroanatomical regions.
- Deployed the trained models as a lightweight web application using Streamlit for real-time inference and public accessibility.

*Technologies: Python, TensorFlow/Keras, CNNs, Transfer Learning, XAI, Streamlit*

[\[GitHub\]](#)   [\[Live Demo\]](#)

## Python Basics: Step-by-Step Learning Repository

Designed and maintained a structured Python learning platform and repository aimed at beginners and early-stage CSE students, covering core programming concepts in a progressive and hands-on manner.

- Organized Python fundamentals into 18 serially numbered modules to ensure a clear and logical learning progression from basic syntax to mini-projects.
- Covered essential topics including variables, data types, conditionals, loops, strings, collections, and basic problem-solving through practical examples.
- Included small interactive programs (calculator, shopping cart, countdown timer) to reinforce learning through implementation.
- Designed the platform to support self-learning, exam preparation, interview fundamentals, and teaching or mentoring activities.

*Technologies: Python*

[\[GitHub\]](#) [\[Learning Website\]](#)

**Restaurant Management Web Application** — Built a Laravel-based full-stack web application with MVC architecture, database integration, and responsive UI using Bootstrap. [\[GitHub\]](#)

## SKILLS

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|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <b>Machine Learning &amp; DL</b> | CNNs, Transformers, Explainable AI (XAI), Transfer Learning  |
| <b>Programming Languages</b>     | Python, C, PHP, JavaScript                                   |
| <b>Machine Learning &amp; DL</b> | CNNs, Transformers, Transfer Learning, XAI, Ensemble Methods |
| <b>Data &amp; Databases</b>      | MySQL, PostgreSQL, MongoDB, SQL                              |
| <b>Web Technologies</b>          | HTML, CSS, Bootstrap, Laravel, React, Node.js                |
| <b>Tools &amp; Platforms</b>     | Git, GitHub, Kaggle, Codeforces, Linux                       |

## REFERENCES

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### Shyla Afroge

Associate Professor, Department of Computer Science & Engineering  
Rajshahi University of Engineering & Technology (RUET)

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